

Education

University of North Carolina at Chapel Hill

Master of Science in Computer Science

August 2024 - May 2026

Indian Institute of Technology Bombay

Bachelor of Technology in Computer Science and Engineering

August 2019 - May 2023

GPA 8.6/10.0

Technical Skills

- **Languages:** Python, C++, C#, JavaScript, TypeScript, SQL, Bash, LaTeX
- **ML and Data:** PyTorch, LightGBM, Transformers, GRU/Mamba sequence models, quantization, pruning, weak supervision, VAEs
- **Systems and Graphics:** Unity, Quest, Apple Metal, OpenGL, SFML, CMake, Linux, profiling, schedulers, automation
- **Web and Tools:** React, Angular, Node.js, MongoDB, Git, REST APIs, data pipelines, technical writing

Experience

Project Research Assistant

Koita Centre for Digital Health, IIT Bombay

September 2023 - June 2024

Mumbai, India

- Built multiomics data augmentation pipeline using **segmented KNN imputation** to reconstruct missing clinical and omics features
- Developed VAE-based sparse branching architecture for multimodal diabetes data, improving downstream test accuracy to **93%**
- Worked across clinical, genomic, and tabular data sources, emphasizing reproducible preprocessing and evaluation

Undergraduate Researcher, Data-Efficient Medical ML

Assistant Prof. Kshitij Jadhav, IIT Bombay

January 2023 - May 2023

Mumbai, India

- Optimized **Spear-DECILE** weak-supervision library for chest x-ray classification using labeling functions and domain rules
- Reduced model training time by **22%** on sparsely labeled medical image data while preserving downstream classification quality
- Trained CNN classifier reaching **0.82 precision** and **0.86 recall**, and documented model behavior for sparse-label settings

Engineering Intern

FinIQ Consulting India Pvt Ltd

May 2022 - July 2022

Pune, India

- Developed Angular client-holdings portfolio dashboard with API integrations for financial product workflows
- Added keyboard event handling and interaction polish to a Foreign Exchange options prototype used in structured product workflows
- Optimized backend operations with multithreading and data parallelization, reducing server-client latency by **2.1x**

Research Intern

IIT Patna

May 2021 - July 2021

Patna, India

- Implemented GAN-based data augmentation pipeline for brain CT tumor imaging experiments
- Evaluated augmented-data classifier reaching **80.2% sensitivity** and **87.6% specificity**

Research and Systems Projects

Adaptive XR Hand-Interaction Governor

Unity, C#, Python, LightGBM, Quest

Spring 2026

UNC Chapel Hill

- Built a Quest/Unity runtime governor for adaptive hand-tracking frequency control across idle, reach, and interaction phases
- Logged OVR metrics, hand telemetry, interaction quality, and phase windows; built repeatable scripts to pull and validate run artifacts
- Trained **LightGBM** intent estimator with **95.19% accuracy** and **0.9285 macro F1**, then deployed flattened C# tree policy
- Combined model policy with dwell and hysteresis protections to prevent unstable mode switching during active hand interaction
- Reduced frame proxy by **8.44%**, app GPU time by **8.56%**, GPU utilization by **7.87%**, and power by **2.92%** while increasing completions by **11.23%**

Brain-to-Text BCI Decoder Efficiency

PyTorch, Mamba, quantization, pruning

Fall 2025

UNC Chapel Hill

- Implemented **GRU**, **Mamba**, and bidirectional Mamba phoneme decoders for the Brain-to-Text 2025 dataset
- Benchmarked PER, latency, memory, pruning tolerance, and energy under FP32 and FP16 inference settings
- Found **FP16** delivered **1.5-3x** speedups with no accuracy loss across supported decoder families
- Achieved **10.61% PER** at **26.5 mJ/sample** with GRU+FP16; evaluated Mamba pruning to a **22M**-parameter footprint
- Built reproducible comparison tables and summaries to separate model quality, runtime cost, and deployment efficiency tradeoffs

Real-Time Boids Scheduling and GPU Acceleration

Fall 2025

C++, Apple Metal, RM/EDF scheduling

UNC Chapel Hill

- Refactored boids simulation into a periodic task set with RM/EDF schedulers, WCET analysis, double-buffered state, and overload control
- Added spatial hashing and multiprocessor scheduling variants to improve scaling under high-boid-count simulation loads
- Implemented Apple Metal compute kernels and profiling harnesses, delivering **7.9x** speedup on 8 threads
- Achieved **5.4x** GPU speedup at 5k boids and raised schedulable capacity from roughly 3k CPU boids to **10k** GPU boids

STLViz: Visual Debugger for C++ Data Structures

Fall 2025

C++20, SFML, CMake

UNC Chapel Hill

- Built C++ visual debugging library wrapping STL-style vector, tree, stack, deque, set, and sorting abstractions while preserving container semantics
- Implemented SFML visualization runtime for algorithm traces, container states, and step-by-step debugging of interview-style programs
- Designed user-study tasks and evaluation protocol to test whether visual traces improved debugging speed and correctness

Image-to-LaTeX OCR

Fall 2024

PyTorch, Transformers, Swin

UNC Chapel Hill

- Built transformer OCR pipeline to translate formula images into LaTeX using a Swin vision encoder and transformer decoder
- Trained on the **latex-formulas** dataset and evaluated outputs with Google BLEU to quantify formula reconstruction quality
- Explored lighter encoder variants and mixture-of-experts ideas for faster formula recognition under constrained model budgets

Additional Engineering Projects

P2P Cryptocurrency Network Simulator

Fall 2022

C++, distributed systems, Ethereum smart contracts

IIT Bombay

- Developed discrete-event simulator for peer-to-peer cryptocurrency networks and modeled selfish and stubborn mining attacks
- Implemented verification, forwarding, mining, fork resolution, and Ethereum smart-contract validation flow

Bash Shell and xv6 Kernel Features

Spring 2021

C, xv6, operating systems

IIT Bombay

- Built shell supporting serial, parallel, and background execution of Linux commands with signal handling
- Implemented lazy page allocation, copy-on-write, pthread-based experiments, round-robin scheduling, and shortest-job-next scheduling

Voxelized Modeling and Animation

Fall 2022

C++, OpenGL, computer graphics

IIT Bombay

- Implemented 3D voxelized modeling and animation software using OpenGL and hierarchical modeling
- Built interface for Bezier space curves and animated track biking with key-frame interpolation

FindMe Proximity Matching App

November 2024

Node.js, MongoDB, mobile backend, HackNC

UNC Chapel Hill

- Led five-person hackathon team building mobile app to match nearby users by shared interests using Wi-Fi proximity signals
- Designed Node.js and MongoDB backend for user tracking, interest matching, and proximity-based discovery workflows

Application Monitoring Dashboard

Spring 2022

React, Telegraf, InfluxDB, systems monitoring

IIT Bombay

- Built web dashboard for real-time application metrics, threshold alerts, and time-series visualization
- Extended monitoring flow across multiple nodes using hosted system telemetry and remote testing infrastructure

Awards and Scholastic Achievements

- National Rank **967** in **JEE Mains** among 1.5 million candidates; recipient of **KVPY** and **NTSE** fellowships
- Member of **Mensa** International, India chapter